Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			**Indicative content** • both ions are free to move in molten state • Al^{3+} ions attracted to negative electrode / move towards negative electrode because opposite charges attract • Al^{3+} ions gain three electrons forming aluminium (atoms) • Al^{3+} ions are reduced • $Al^{3+} + 3e^- \rightarrow Al$ • O^{2-} ions attracted to positive electrode / move towards positive electrode because opposite charges attract • O^{2-} ions lose two electrons forming oxygen (atoms) • two O^{2-} ions lose (four) electrons forming oxygen (molecules) • O^{2-} ions are oxidised • $2O^{2-} - 4e^- \rightarrow O_2$ **5-6 marks** Clear understanding of movement of ions and gain/loss of electrons; good attempt at equations *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.*	6			6		